App Flow

Step 1: Onboarding & Ingestion

- Sigin in and sigin page.
  User opens the app and selects UI language (Urdu/English).
- User navigates to the "Doc Vault" and uses the camera to scan a marksheet or transcript.
- *Agent Action:* Ingestor Agent extracts Name, CGPA, Field, and University from the image.

Step 2: Insight & Gap Detection

- *Agent Action:* Matcher Agent processes the data and identifies top matches (e.g., HEC Need-Based, Fulbright).
- *Agent Action:* Gap Detector Agent analyzes requirements and highlights missing items (e.g., "Missing GRE score, 2 recommendation letters").

Step 3: Action Execution (Simulation)

- The system executes an action chain:
  - *Action 1:* Drafts and simulates sending an email to a university professor requesting a recommendation, pre-filled with student details.
  - *Action 2:* Auto-fills a mock scholarship application form with extracted data.
  - *Action 3:* Creates a deadline tracking calendar entry.

Step 4: Outcome & Trace Visibility

- The user is redirected to the Readiness Dashboard.
- The dashboard displays a clear "Before vs. After" state (e.g., Before: 0 scholarships tracked. After: 3 matches, 1 email sent, form 60% complete).
- An "Agent Trace Panel" is visible on the screen, showing the exact reasoning logs of the Antigravity engine to the judges.

4. UI/UX Design Brief

Visual Style:

- **Framework:** Material 3 UI implementation for clean, native Android components.
- **Accessibility:** Must include an Urdu UI toggle, Urdu text-to-speech voice output, and support for mixed-language/Roman Urdu inputs.

Key Screens:

- **Camera Ingestion Screen:** Overlay guiding the user to snap clear photos of documents. Immediate visual feedback showing data extraction (OCR).
- **Gap Radar View:** Red/Yellow/Green urgency indicators showing the priority sequence for missing documents (e.g., Domicile takes 2 weeks = Red).
- **Simulation & Outcome Screen:** Crucial for hackathon scoring. Must visibly show the "Before" and "After" state of the system.
- **Antigravity Debug View:** A dedicated, visible panel or toggle specifically for judges to read the Antigravity logs (workplan, decisions, tool calls) live during the demo.

5. Backend Schema
The backend will rely on Firebase Firestore for rapid prototyping.
Collection 1: Users

- user_id (String, Primary Key)
- extracted_profile (Map: name, cgpa, university, field_of_study)
- language_preference (String: "ur" or "en")
- overall_readiness_score (Number)

Collection 2: Scholarships **(Hardcoded Pakistan DB)**

- scholarship_id (String)
- name (String: e.g., "HEC Need-Based", "PEEF")
- type (String: "Local", "International")
- eligibility_criteria (Map: min_cgpa, required_degree_level)
- required_documents (Array of Strings)
- deadline_date (Timestamp)

Collection 3: UserDocuments

- doc_id (String)
- user_id (Reference to Users)
- doc_type (String: "Marksheet", "Transcript", "Passport")
- completeness_score (Number)
- extracted_text (String)

Collection 4: ActionTraces **(For Hackathon Logging)**

- trace_id (String)
- user_id (Reference to Users)
- antigravity_logs (Array of Maps containing: step, reasoning, tool_called, result, fallback_triggered)
- timestamp (Timestamp)